

Question number	Answer	Additional guidance	Mark
8(a)	An explanation that includes four of the following points: • (because a large spleen) can {store / release} {more / a lot of} {red blood cells / oxygen} (1) • oxygen {dissociates from the haemoglobin / diffuses out of the red blood cells} (1) • (delivering / providing) oxygen (to the cells) for aerobic respiration (1) • allowing the seals to stay {underwater / dive} longer (1) • credit an advantage of this (1)	DO NOT ACCEPT {store / release} of blood ACCEPT muscles can contract for longer e.g. prevent buildup of lactic acid seals can escape predators seals can find food	(4)

Question number	Answer	Mark
8(b)(i)	The only correct answer is B <i>A is incorrect because the mean is the sum of all values divided by the number of values</i> <i>C is incorrect because mode is the most common value</i> <i>D is incorrect because standard deviation is a measure of variance</i>	(1)

Question number	Answer	Additional guidance	Mark
8(b)(ii)	A description that includes two of the following points: • {median / middle} size of spleens of Saluan people is smaller (than the Bajau people) (1) • (some) Bajau people have larger spleens than any of the Saluan people (1) • there is a greater {range / spread / variation} in the sizes of spleens in the Bajau people than the Saluan people (1)	ACCEPT converse ACCEPT some Saluan have larger spleens than the Bajua	(2)

Question number	Answer
8(c)	Aspect 1 : how spleens became enlarged • mutation may have occurred • in gene coding for {spleen size / ability to store more red blood cells} • spleen size maybe polygenic / multi-allelic • so wide range of spleen sizes • selection pressure for larger spleens • {increased chance of survival / more food / more oxygen in body} acted as a selection pressure Aspect 2 : advantages of large spleens • large spleens allowed the Bajau people to dive for longer • as more oxygen was supplied to their cells • when they were holding their breath • enabling them to spend more time hunting under water / not returning surface • and therefore catching more food • increasing their chances of survival Aspect 3 : how the allele frequency increased • natural selection • those who survived were more likely to reproduce • passing the alleles onto their offspring • offspring more likely to survive • repeated over many generations • increasing the frequency of the allele in the population Aspect 4 : interactions with divers and non-divers / lack of interactions with Saluan people • Bajau people did not reproduce with Saluan people • because the Bajau people lived in house boats / Saluan people lived on the main land • therefore the alleles not introduced into this population of people • or if it was it became diluted • diving and non-diving Bajau people reproduced together • therefore the allele for large spleens was present in both divers and non-divers

			Additional guidance
level 0	0	No awardable content	
level 1	1-2	An explanation may be attempted but with limited interpretation or analysis of the scientific information and with a focus on mainly just one piece of scientific information. The explanation will contain basic information, with some attempt made to link knowledge and understanding to the given context.	Simple comments / explanation 1 mark = simple explanation of one aspect 2 marks = simple explanation of two aspects
level 2	3-4	An explanation will be given, with occasional evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows some linkages and lines of scientific reasoning, with some structure.	Some extended explanation 3 marks = extended explanation of one aspect 4 marks = at least two aspects covered which includes extended explanation of one of the aspects
level 3	5-6	An explanation is made that is supported throughout by sustained application of relevant evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows a well-developed and sustained line of scientific reasoning, which is clear and logically structured.	Extended explanation clearly linked to size of spleen 5 marks = extended explanation of two aspects 6 marks = coherent extended explanation of three aspects

